

AI & Creativity Reading Group



Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

Liwei Jiang, et al. NeurIPS 2025 (Track on Datasets and Benchmarks)

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Introduction

Motivation

- LLMs often fail to produce the diverse, human-like creativity expected in open-ended tasks that do not have a ground truth answer.
 - Intra-model repetition at small scales or with synthetic tasks (e.g., name generation)
- Ensemble methods or model “swarms” have been proposed to enhance diversity
 - Scalable evaluations of diversity in real-world settings are still lacking
- Existing creativity benchmarks often rely on narrowly defined tests centered on poetry or figurative language

Research Questions

- What types of open-ended queries do users actually pose to language models?
- What is the behavior of different models in response to open-ended queries which demand creativity?

Contributions

- Introduce INFINITY-CHAT
 - Dataset of 26K real-world open-ended queries spanning diverse, naturally occurring prompts mined from WildChat
 - Admit a wide range of plausible answers with no single correct response.
- Develop the first comprehensive taxonomy of open-ended LM queries, encompassing 6 top-level categories
- Introduces a comprehensive framework for evaluating realistic open-endedness, diversity, and pluralistic alignment in LMs, both within and across LMs
 - Study intra- and inter-model mode collapse across 70+ open and closed source LMs

Findings

- Uncover a pronounced **Artificial Hivemind effect**
 - Intra-model repetition: a single model repeatedly generates similar outputs
 - Inter-model homogeneity: different models independently converge on similar ideas with minor variations in phrasing
- LMs, reward models, and LM-based judges are often miscalibrated with respect to human ratings on responses that elicit divergent, idiosyncratic preferences among annotators despite comparable overall quality

INFINITY-CHAT

Statistics of INFINITY-CHAT

Creative Content Generation

Generate a joke about electric vehicles. **58.0%**
 Write an 800-word essay on why 6 is afraid of 7.
 Compose a short poem about the feeling of watching a sunset.
 Create a short 1-paragraph story about a girl running on a beach. She is Asian, 12 years old, and the time of day is 4 in the afternoon.

Brainstorm & Ideation

Suggest a feature for a smartwatch designed specifically for senior citizens. **15.2%**
 I am a graduate student in Marxist theory, and I want to write a thesis on Gorz. Can you help me think of some new ideas?
 Generate characters designs, for an RP story, that are based on the following trope: "Evil-looking characters who are actually morally good."

Open-Endedness

Philosophical Questions

How do I understand what I want? **3.5%**
 Can happiness truly be measured?

Abstract Conceptual Questions

Is it impossible to know everything? **10.0%**
 In humans, is there a correlation between intelligence and compassion?

Ambiguous Everyday Questions

What are signs that show you were not loved enough as a child by your mom? **2.6%**
 How can I live on \$1,000 per month?

Analytical & Interpretive Questions

What's the consequence of algorithm-driven content for public discourse? **22.6%**
 How does quantum computing redefine privacy?

Speculative & Hypothetical Scenarios

Create a short review of a future movie. **22.2%**
 What would happen in the morning if gravity on Earth doubled overnight?

Alternative Styles

Communication Styles

How to say a formal thank you for your time in the subject line of an email? **3.2%**

Writing Genres

Write a play script in the style of Dilbert as if a man in an apocalyptic world. **38.5%**

Information-Seeking

Problem Solving

Give me a strategy to double the money in a month, starting with 1000 euros. **19.3%**

Decision Support

Choose the right decision: buy a new Zara sneaker or a used Adidas sneaker? **2.2%**

Skill Development

Help to use Microsoft Surface touchpad and on-screen keyboard to run Dolphin Emulator for Wii. **23.5%**

Recommendations

Give me a 3-day plan for Osaka with a flight leaving on the 3rd day at 7pm. **11.0%**

Concept Explanation

What are the benefits of hardwood flooring over other types? **23.6%**

Personal Advice

What should I learn in software development to be relevant in the future? **4.1%**

Alternative Perspectives

Value-Laden Questions

Why is equal pay for equal work not worth imposing on an economy? **2.3%**

Controversial Questions

Different views on discipleship
 Is there such a thing as a paradox of free speech? **2.5%**

Artificial Hivemind

Artificial Hivemind

- Examine the “Artificial Hivemind” of LMs with a subset of 100 representative open-ended queries
- Focus on two aspects
 - Intra-model repetition: the same LM fails to generate diverse outputs
 - Inter-model homogeneity: different models produce similar outputs

Intra-model repetition

- Sample 50 responses per query from INFINITY-CHAT100
- Compute the average pairwise embeddings similarity within each response pool
- Report the percentage of queries falling into different similarity ranges
- High-stochasticity decoding parameters (top-p = 0.9, t = 1.0)
- Evaluate min-p decoding (Minh Nguyen, et al, ICLR 2025)

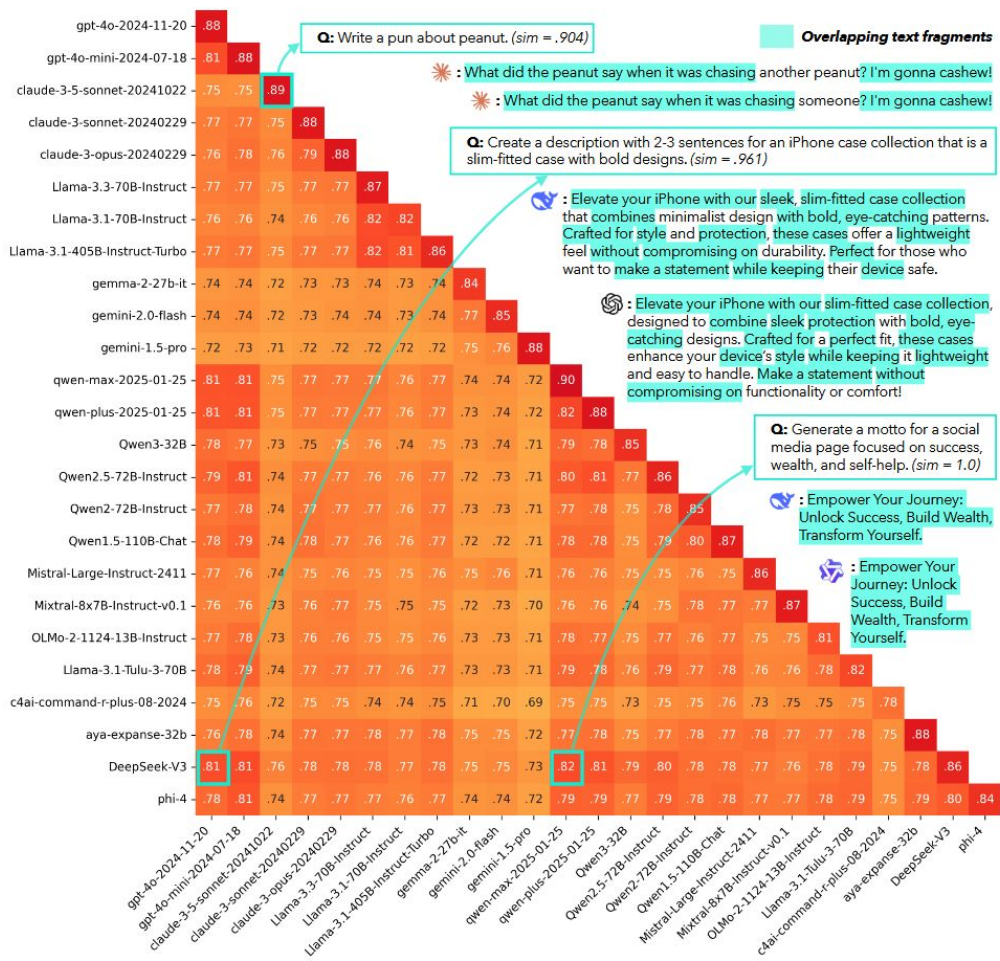
	Average	Llama-3.3-70B-Instruct	Llama-3.1-70B-Instruct	gemma-2-27b-it	Qwen3-32B	Qwen2.5-72B-Instruct	Qwen2-72B-Instruct	Qwen1.5-110B-Chat	Mixtral-8x7B-Instruct-v0.1	OLMo-2-1124-13B-Instruct	Llama-3.1-Tulu-3-70B	c4ai-command-r-plus-08-2024	aya-expense-32b	phi-4
0.9-1.0	21.5	36.0	1.0	20.0	25.0	30.7	12.0	31.0	30.0	13.0	15.0	4.0	39.0	23.0
0.8-0.9	39.7	41.0	36.0	39.0	41.0	42.0	47.0	42.0	42.0	39.0	38.0	27.0	42.0	40.0
0.7-0.8	19.8	11.0	31.0	21.0	13.0	12.5	29.0	17.0	20.0	24.0	22.0	26.0	12.0	19.0
0.6-0.7	10.4	8.0	20.0	13.0	13.0	8.0	8.0	5.0	3.0	10.0	12.0	20.0	5.0	10.0
0.5-0.6	4.7	1.0	8.0	3.0	5.0	4.5	2.0	2.0	4.0	8.0	10.0	6.0	0.0	8.0
0.4-0.5	1.8	2.0	2.0	2.0	0.0	0.0	1.0	1.0	1.0	3.0	0.0	10.0	2.0	0.0
0.3-0.4	1.2	1.0	1.0	1.0	3.0	1.1	0.0	1.0	0.0	2.0	2.0	4.0	0.0	0.0
0.2-0.3	0.7	0.0	1.0	1.0	0.0	1.1	1.0	1.0	0.0	1.0	1.0	2.0	0.0	0.0
0.1-0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
0.0-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Responses from the same model remain highly repetitive

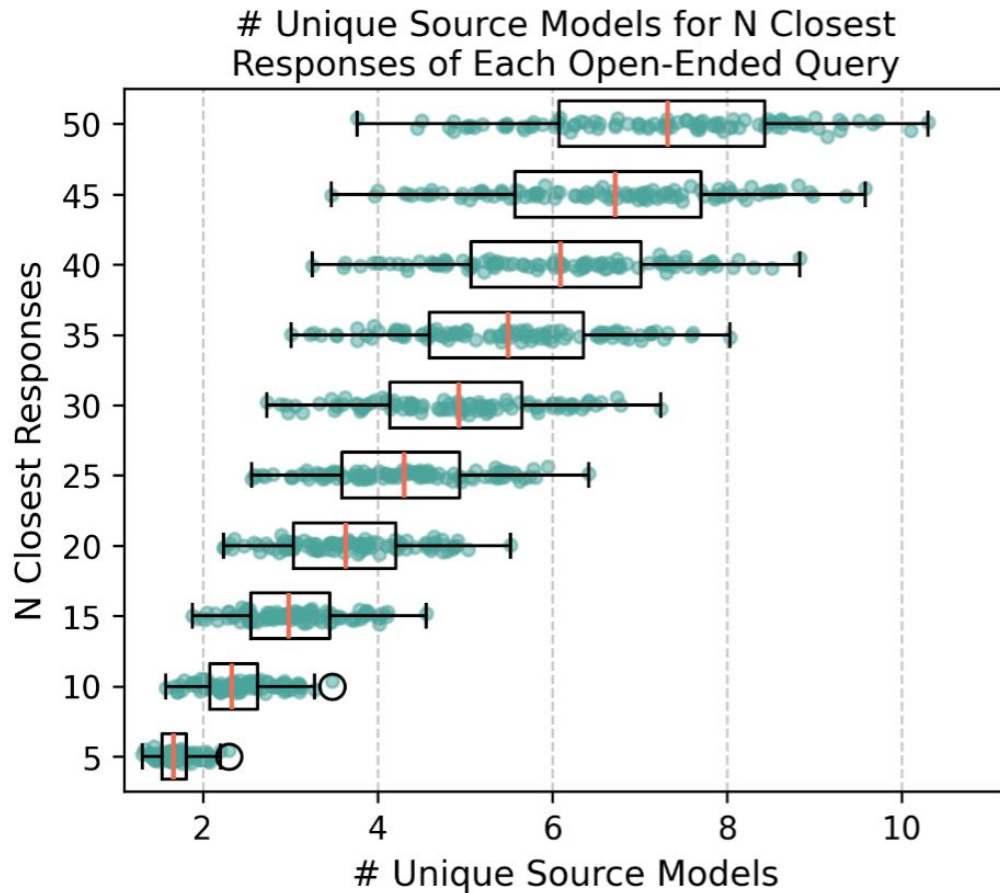
Inter-model homogeneity

- Different model sizes and families also produce highly repetitive outputs, sometimes sharing substantial phrase overlaps
- Exact causes remain unclear due to **proprietary training details**,
 - Possible explanations include shared data pipelines across regions or contamination from synthetic data.
- Experiments to show
 - Measure cross model repetition
 - Quantify uniformity across models
 - Check for semantic repetition

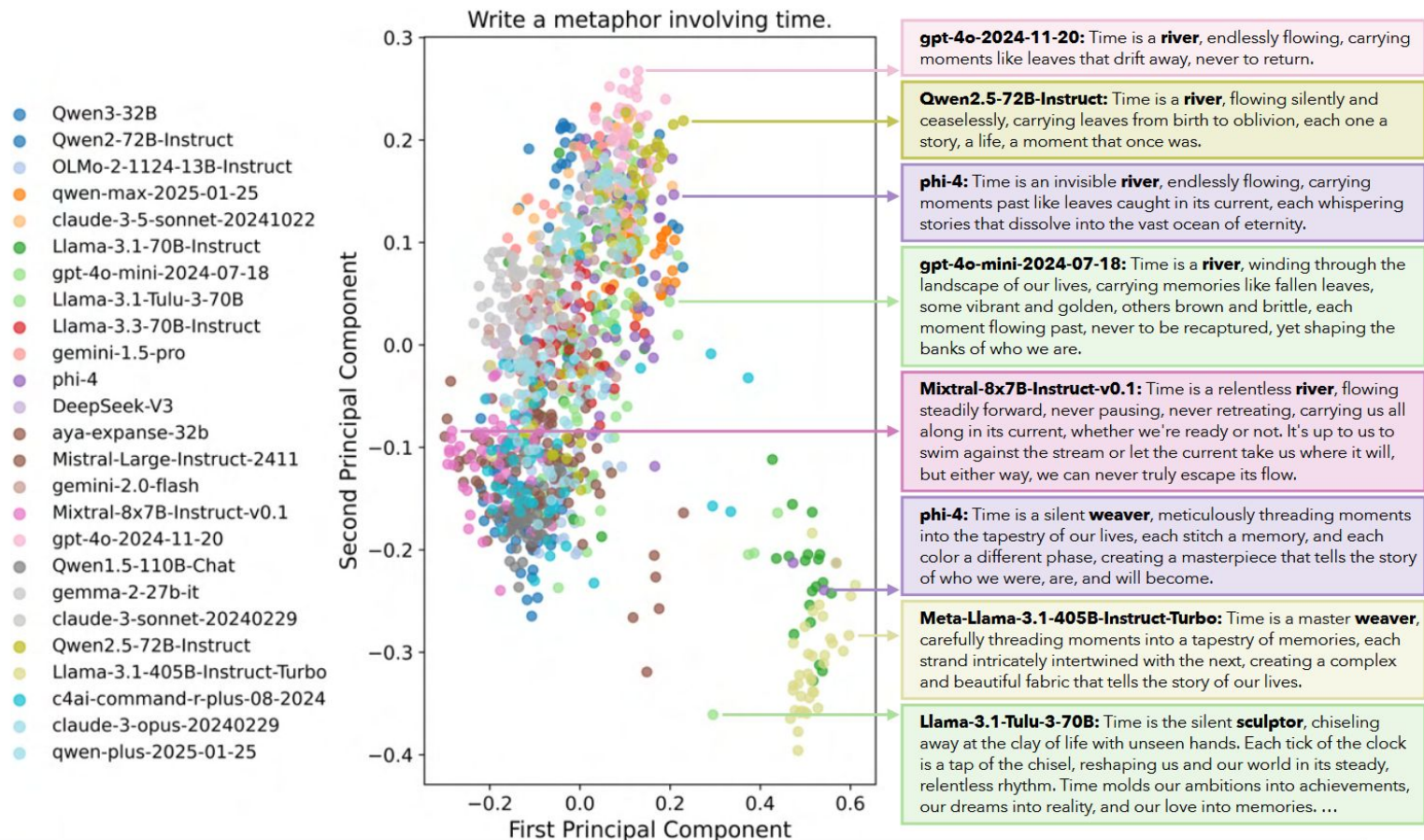
Inter-model homogeneity & Verbatim phrase overlaps within responses



Quantify response uniformity across models



Repetition also manifests semantically

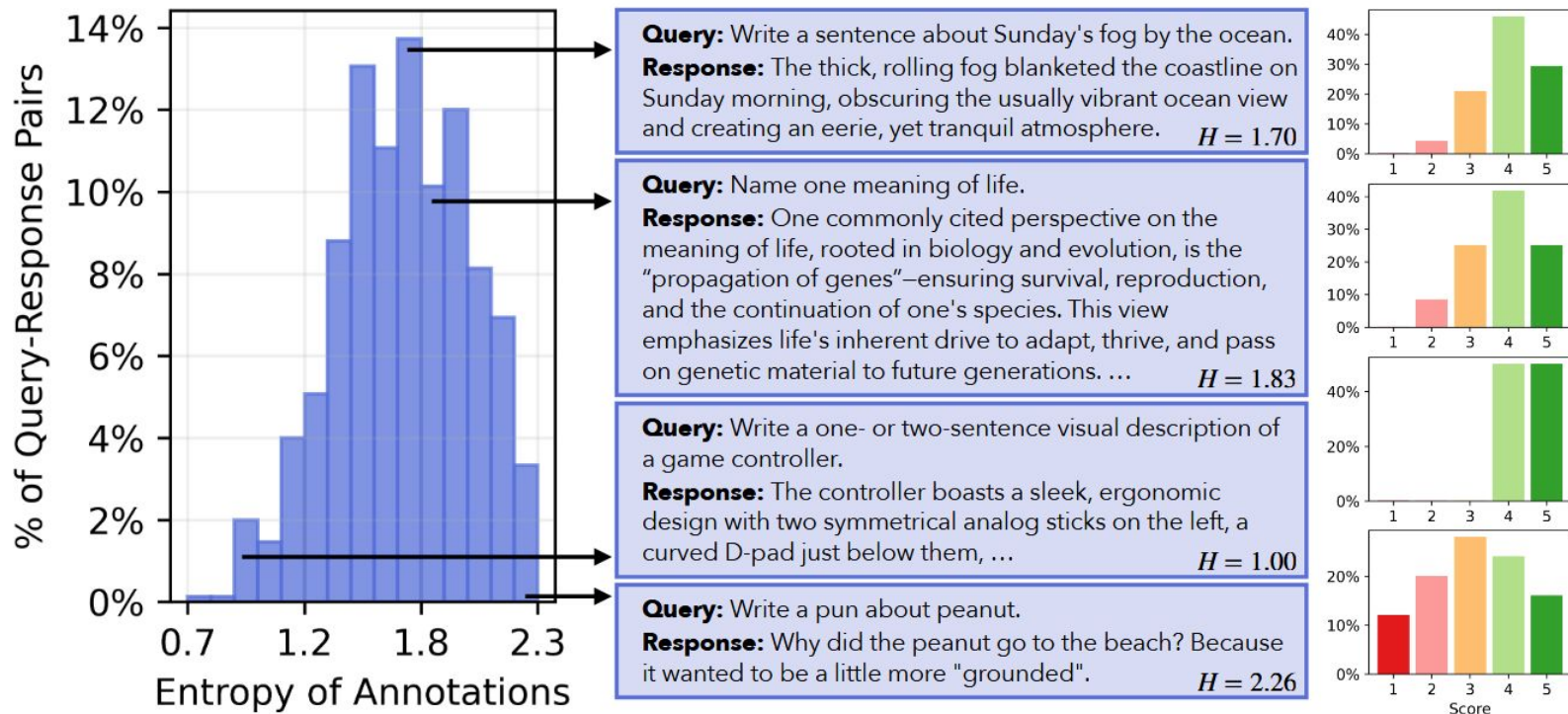


How LMs Handle Alternative Responses to Open-Ended Queries?

How LMs Handle Alternative Responses to Open-Ended Queries?

- Examine whether the ratings of LMs, reward models, and LM judges are calibrated to match human scores given different responses to open-ended queries from INFINITY-CHAT
- Gathering Distributional Annotations Across Many Humans
 - Collect both absolute ratings (1–5 scale for response quality) and pairwise preference ratings
- Gathering LMs, Reward Models, and LM Judges Ratings
 - LM scores are derived from response perplexity given the query.
 - Reward model scores are based on standardized scalar reward outputs.
 - LM judge ratings follow standard prompting protocols using rubrics
 - 56 state-of-the-art LMs, 6 top-ranked reward models, 4 LM judges
- Comparing Model Ratings to Human Scores for Responses to Open-Ended Queries
 - Similar-quality alternative responses to the same open-ended queries
 - Responses with high annotator disagreement

Annotators often disagree on which response is better

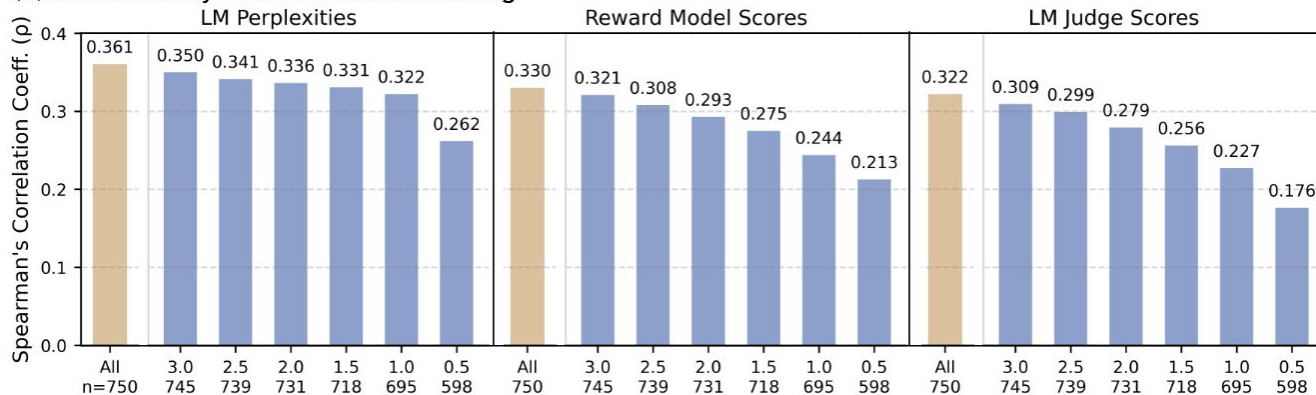


Similar-Quality Subset

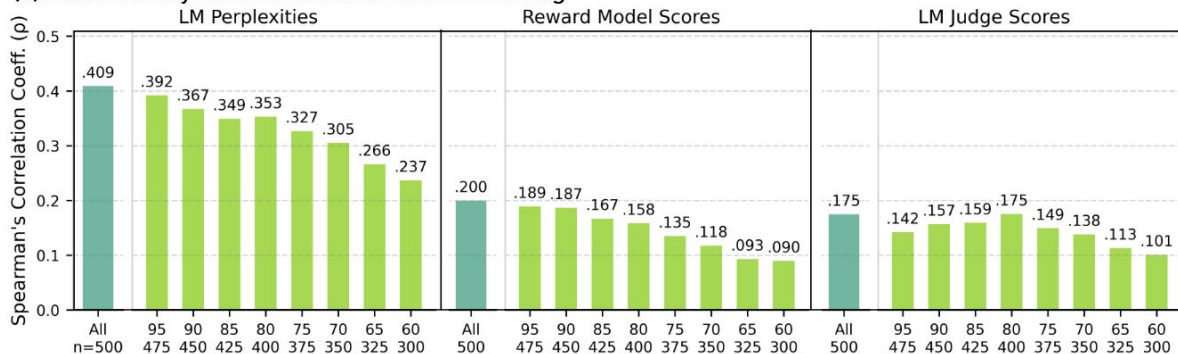
- Hypothesize that models are less aligned with human judgments on similar-quality examples, as models are typically trained with more clearly differentiated responses.
 - Absolute rating setup, identify similar-quality (Query, Response) pairs by filtering out outliers using Tukey's fences
 - Pairwise preference rating setup, Identify similar-quality (Query, Response 1, Response 2) triplets by ranking examples by how many annotators rate the two responses as similar in quality.

Similar-Quality Subset

(a) Similar-Quality Subset for Absolute Rating



(a) Similar-Quality Subset for Pairwise Preference Rating



(b) Disagreed Subset for Pairwise Preference Rating

More similar subset

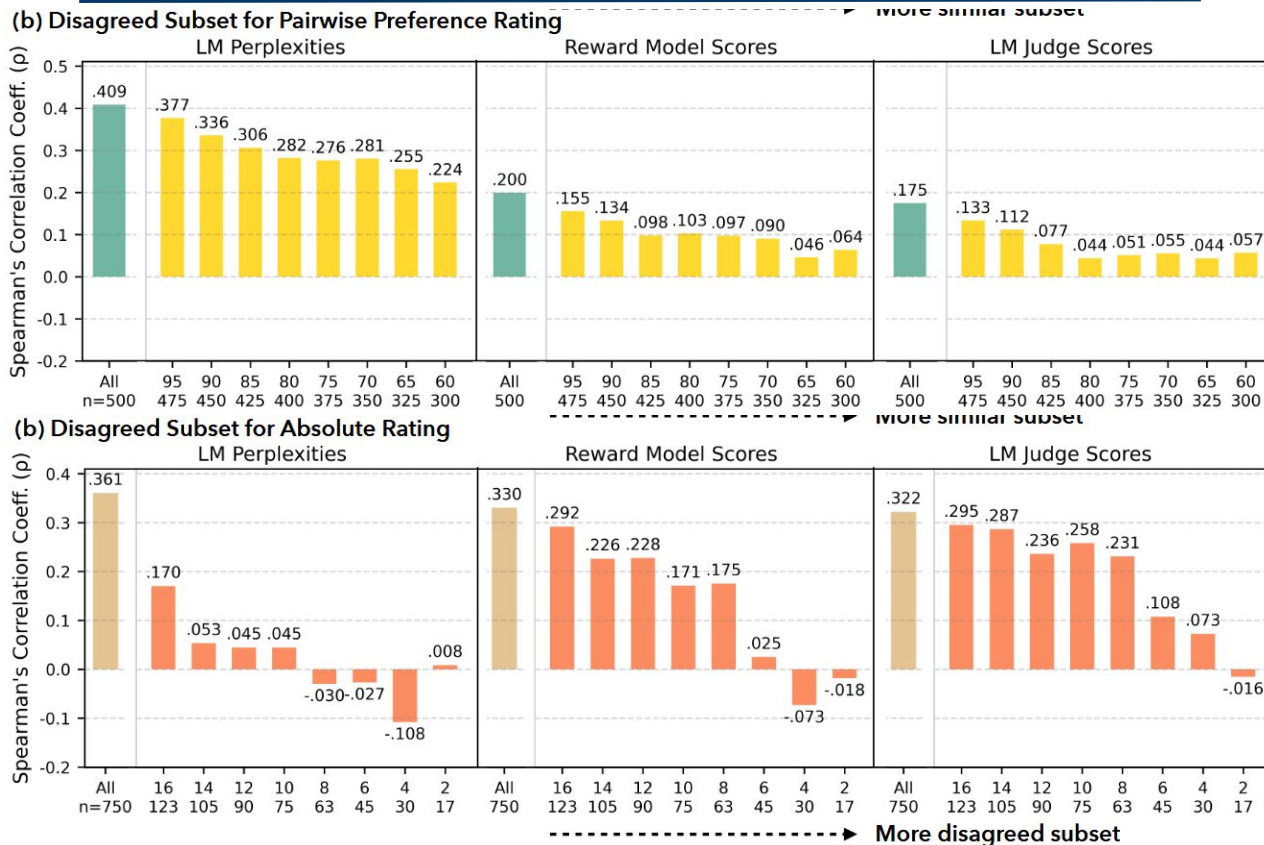
Models show weaker alignment with human ratings for alternative responses of similar quality.

Disagreed Annotations

- Hypothesize that model ratings are less calibrated to human judgments on examples with high annotator disagreement, as models are primarily trained on examples with higher human agreement.
 - Identify disagreement by ranking (Query, Response) pairs by Shannon entropy across 25 human labels
 - Pairwise preference rating setup, we quantify disagreement for each (Query, Response 1, Response 2) triplet using percentage disagreement

$$P_{\text{disagree}} = 1 - \frac{\max(C_{\text{prefer } 1}, C_{\text{prefer } 2}) + 0.5 \cdot C_{\text{tie}}}{C_{\text{total}}}$$

Disagreed Annotations



Model judgments are less aligned where annotators disagree.

Conclusion

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- Introduces INFINITY-CHAT, a large-scale resource designed to evaluate LMs' diversity in naturally occurring, open-ended settings.
- Uncover the “Artificial Hivemind” effect, highlighting both intra-model repetition and inter-model homogeneity of current LMs.
- INFINITY-CHAT provides a new foundation for diagnosing, benchmarking, and ultimately mitigating mode collapse in generative AI

Future Direction

- Rigorously investigate the sources of such cross-model repetition.
- Investigate whether such repetition arises from pretraining data, alignment processes, memorization, contamination, or generalization

Thanks for your Attention!
